

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct length $\frac{w}{\sqrt{2}} = \frac{\sqrt{2}w}{2}$ ACF	AO1.1b	B1	$\frac{w}{\sqrt{2}}$
(b)	Models the lengths as a geometric sequence	AO3.3	M1	$a = w$ and $r = \frac{1}{\sqrt{2}}$
	Finds the sum to infinity provided their $r < 1$	AO1.1a	M1	$S_{\infty} = \frac{w}{1 - \frac{1}{\sqrt{2}}}$
	Uses their model to obtain the correct sum in terms of w	AO3.4	A1	$\approx 3.41w < 3.5w$
	Compares their sum with $3.5w$	AO2.4	E1	
(c)	Explains that the model would have to include an additional 3 mm for each tile	AO3.5c	E1	The total length will now include an additional 3 mm for each tile. The total length will not have an upper limit.
	Explains that the total length will not have an upper limit Or The total length may now exceed $3.5w$	AO3.5a	E1	
				Total 7 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds a difference between 2 terms	AO3.1a	M1	$3e^p - 5 = 5 - 3e^{-p}$ (*)
	Forms an equation using two differences	AO3.1a	M1	$3e^p - 10 + 3e^{-p} = 0$ $3e^{2p} - 10e^p + 3 = 0$
	Forms a quadratic	AO1.1a	M1	$e^p = \frac{1}{3}, 3$

	equation in e^p			$p = \ln \frac{1}{3}, \ln 3$
	Obtains a correct quadratic equation	AO1.1b	A1	ALT to (*) $2(5 - 3e^{-p}) = 3e^p - 3e^{-p}$ Or $2(3e^{-p} - 5) = 3e^p - 3e^{-p}$
	Obtains 2 correct solutions for e^p from 'their' quadratic FT only applies if previous mark has been awarded	AO1.1b	A1F	
	Obtains final answers in an exact form FT applies if previous mark has been awarded	AO2.2a	A1F	
(b)	Finds a ratio between two consecutive terms (no requirement to use a and r)	AO3.1a	M1	Assume it is possible that $3e^{-q}$, 5 and $3e^q$ are three consecutive terms of a geometric sequence
	Compares two ratios (could be ratios of successive terms, no requirement to use a and r)	AO3.1a	M1	$a = 3e^{-q}, ar = 5, ar^2 = 3e^q$ $\frac{ar}{a} = \frac{5}{3e^{-q}} \Rightarrow r = \frac{5e^q}{3}$ $\frac{ar^2}{ar} = \frac{3e^q}{5} \Rightarrow r = \frac{3e^q}{5}$
	Identifies a contradiction	AO2.1	R1	$\frac{5}{3e^{-q}} = \frac{3e^q}{5} \Rightarrow 25 = 9$
	Draws a conclusion about k	AO2.4	R1	This is a contradiction therefore $3e^{-q}$, 5 and $3e^q$ cannot form three consecutive terms of a geometric sequence.
Total 10 marks				

Q3.

(a) (i) $r = \frac{294}{420} = 0.7$

AG. Accept any valid justification to the given answer

B1

1

(ii) $\{S_{\infty} = \} \frac{a}{1-r} = \frac{420}{1-0.7}$
 $\frac{a}{1-r}$ used

M1

$$\{S_{\infty}\} = 1400$$

1400 NMS mark as 2 / 2 or 0 / 2

A1

2

(iii) n th term = $600 \times (0.7)^n$

If not B2 award B1 for $420 \times (0.7)^{n-1}$ OE

B2

2

(b) (i) $\{u_n\} = 248 - 8n$

Accept ACF

B1

1

(ii) $u_k = 0 \Rightarrow 8k = 248$

$248 - 8k = 0$ OE e.g. $240 + (k - 1)(-8) = 0$
ft if no recovery, on c's (b)(i) answer

M1

$$k = 31$$

A1

$$\sum_{n=1}^k u_n = 240 + 232 + \dots + 0 = \frac{k}{2} [240 + 0]$$

For $\frac{k}{2} [240 + 0]$ or for $\frac{k}{2} [c's u_1 + 0]$

OE e.g. $\frac{k}{2} [2 \times c's u_1 + (k - 1)(-8)]$

M1

$$\sum_{n=1}^k u_n (= 15.5 \times 240) = 3720$$

A1

4

[10]

Q4.

(a) 20

20

B1

1

(b) $\{S_{\infty} = \frac{a}{1-r} = \frac{80}{1-\frac{1}{2}}\}$
 $\frac{a}{1-r}$ used with $a = 80$ and $r = 0.5$ OE

M1

$\{S_{\infty} = \} 160$

NMS 160 gets 2 marks unless rounding seen

A1

2

(c) $\{S_{12} = \frac{80(1-r^{12})}{1-r} = 160(1-0.5^{12})\}$
 $\frac{80(1-r^{12})}{1-r}$ seen (or used with $r = 0.5$ OE)

$= 159.96(0937.) = 159.96$ to 2dp

Condone > 2dp

A1

2

[5]

Q5.

(a) $u_1 = 12$

CAO Must be 12

B1

$u_2 = 48 \times \frac{1}{16} = 3$

If not correct, ft on c's $u_1 \times \frac{1}{4}$

B1F

2

(b) $r = \frac{1}{4}$

Only ft on $r = (c's u_2) \div (c's u_1)$ if $|r| < 1$. Answers may be in equivalent fraction form or exact decimal form. If other notation used award the mark if correct or ft value confirmed in (c)

B1F

1

(c) $(S_{\infty} =) \frac{u_1}{1-r} = \frac{12}{1-\frac{1}{4}}$

Use of $\frac{a}{1-r}$ ft on c's u_1 and c's r in (a) and (b) if not recovered, provided $|r| < 1$

M1

= 16

If not 16, ft on c's u_1 and c's r in (a) and (b) provided $|r| < 1$

A1F

2

(d) $\sum_{n=4}^{\infty} u_n = S_{\infty} - \sum_{n=1}^3 u_n$

OE eg RHS $S_{\infty} - (u_1 + u_2 + u_3)$

M1

$u_3 = \frac{3}{4}$ (or $\sum_{n=1}^3 u_n = \frac{12(1-0.25^3)}{1-0.25}$)

Either result, or better eg $\sum_{n=1}^3 u_n = 15.75$

B1

$\sum_{n=4}^{\infty} u_n = 0.25$

NMS scores 0 / 3

A1

SC For c's scoring 0 / 3 in (d); Award B1 to candidates

who used $S_{\infty} - S_4$ for $\sum_{n=4}^{\infty} u_n$ and obtained the answer $\frac{1}{16}$ OE

3

Alternative

$(\sum_{n=4}^{\infty} u_n = \frac{u_4}{1-r})$

(M1)

$(u_4 = \frac{3}{16} (= 0.1875))$

(B1)

$$\left(\sum_{n=4}^{\infty} u_n = \frac{3}{16} \div \frac{3}{4} = \frac{1}{4} \right)$$

(NMS scores 0 / 3)

(A1)

(3)

[8]

Q6.

(a) (i) $ar^2 = 36$; $ar^5 = 972$;

For $ar^2 = 36$ or $ar^5 = 972$ or for seeing $36r^3 = 972$

M1

$$r^3 = \frac{972}{36} (= 27) \Rightarrow r = 3$$

CSO AG Full valid completion.

A1

2

(ii) $a \times 3^2 = 36$

OE. PI

M1

$$a = 4$$

Correct answer without working scores the two marks

A1

2

(b) (i) $\sum_{n=1}^{20} u_n = S_{20} = \frac{a(1-r^{20})}{1-r}$

OE

M1

$$= \frac{4(1-3^{20})}{-2} = -2(1-3^{20}) = 2(3^{20}-1)$$

CSO AG Be convinced

A1

2

(ii) $u_n = a \times 3^{n-1}$

Seen or used

B1

$$= 4 \times 3^{n-1} > 4 \times 10^{15} \Rightarrow 3^{n-1} > 10^{15}$$

$$(n - 1) \log 3 (>) \log 10^{15}$$

Or finds values of u_n for appropriate adjacent integer values of n so that u_n 's are either side of 4×10^{15}

M1

$$n - 1 > \frac{15}{\log_{10} 3}; n - 1 > 31.4...$$

$(n > 32.4... \text{ and } n \text{ is an integer so least value of } n \text{ is})$
 $n = 33$

CSO

A1

3

[9]

Q7.

$$(a) \quad \{S_{\infty} = \} \quad \frac{a}{1-r} = \frac{12}{1-\frac{3}{8}}$$

$$\frac{a}{1-r} \quad \text{used}$$

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M1

$$\{S_{\infty} = \} 19.2$$

19.2 OE NMS mark as 2/2 or 0/2

A1

2

$$(b) \quad \{6\text{th term} = \} ar^{6-1}$$

Stated or used

M1

$$= 12 \times \left(\frac{3}{8}\right)^5 = 2 \times 2 \times 3 \times \left(\frac{3}{2 \times 2 \times 2}\right)^5$$

Changing 8 and 12 in correct expression to correct products/powers of 2 and 3

m1

$$= \frac{2 \times 2 \times 3 \times 3^5}{(2^3)^6} = \frac{2^2 \times 3^6}{2^{18}} = \frac{3^6}{2^{16}}$$

AG Be convinced

A1

3

(c) (i) $\{u_n\} = 12x\left(\frac{3}{8}\right)^{n-1}$
 OE. eg $32(3/8)^n$

B1

1

(ii) $\log u_n = \log 12 + \log\left(\frac{3}{8}\right)^{n-1}$

$$\log u_n = \log 12 + (n-1)\log\left(\frac{3}{8}\right)$$

$$\log u_n = \log 12 + (n-1)[\log 3 - \log 8]$$

Log laws

$$\log(PQ) = \log P + \log Q ;$$

$$\log\left(\frac{P}{Q}\right) = \log P - \log Q$$

$$\log(P)^k = k \log P$$

Using (c)(i) and taking logs: one log law used correctly, on a correct expression for u_n

M1

a second different log law used correctly,
 indep of prev M error and ft on cand's (c)(i)
 provided cand's u_n expression has a power
 involving n .

M1

$$\log u_n = \log 3 + 2 \log 2 + (n-1)[\log 3 - 3 \log 2]$$

A third different log law used correctly (or
 equivalent valid step) to reach a correct RHS
 whose terms are all multiples of $\log 2$ and $\log 3$.
 Dep on both prev two Ms

m1

$$\log u_n = n \log 3 - 3n \log 2 + 5 \log 2$$

$$\log_a u_n = n \log_a 3 - (3n - 5) \log_a 2$$

CSO AG Be convinced, no slips although we will condone the absence of the bases a even in the final line.

A1

4

[10]

Q8.

(a) (i) $\{S_\infty = \} \frac{a}{1-r} = \frac{10}{1-r}$

$$\frac{a}{1-r} \text{ used}$$

M1

$$\frac{10}{1-r} = 50 \text{ so } 1-r = \frac{10}{50} \Rightarrow r = \frac{4}{5}$$

AG Condone verification with the correct final statement but be convinced.

A1

2

(ii) 2nd term = ar

ar stated or used for the 2nd term. PI by ans'8'

M1

$$= 8$$

A1

2

(b) (i) 4th term = $a + 3d$; 8th term = $a + 7d$

Uses $a + (n - 1)d$ correctly at least once

M1

$$a + 3d = 10, \quad a + 7d = 8$$

Both eqns. correct ft on c's (a)(ii) OE eg
 $8 = 10 + 4d$

A1F

$$\Rightarrow 4d = -2 \Rightarrow d = -0.5$$

OE fraction.

A1

3

(ii) $a + 3(-0.5) = 10$

*An appreciation that a is required in
(b)(ii) and a valid method to find a
anywhere or PI if $a = 11.5$ seen/used*

M1

$\Rightarrow a = 11.5$

A1F

*Ft on c 's non-zero value for d
ie using $a = 10 - 3d$ or $a = c$'s $8 - 7d$.
[c 's 8 is candidate's answer to (a)(ii)]*

$$\sum_{n=1}^{40} u_n = S_{40} = \frac{40}{2} [2a + (40 - 1)d]$$

$$\frac{40}{2} [2a + (40 - 1)d] \quad \text{OE}$$

M1

$= 70$

A1

4

[11]

Q9.

(a) (i) $ar = 375; ar^4 = 81$

For either OE or PI by next line

B1

$\Rightarrow 375r^3 = 81$

Elimination of a OE

M1

$$r^3 = \frac{81}{375} = \frac{27}{125} = 0.216 \Rightarrow r = 0.6$$

*CSO AG Full valid completion
SC: Clear explicit verification, with
statement max B1 out of 3. (If considers
uniqueness then 3 is possible)*

A1

3

(ii) $0.6a = 375$
OE; PI

M1

$$a = 625$$

A1

2

(b) $\frac{a}{1-r} = \frac{a}{1-0.6}$
 $\frac{a}{1-r}$ **used** with $| \text{value of } r | < 1$

M1

$$S_{\infty} = \frac{625}{0.4} = 1562.5$$

ft on cand's value for a.... ie $2.5 \times a$

A1F

2

(c) $\sum_{n=6}^{\infty} u_n = \sum_{n=1}^{\infty} u_n - \sum_{n=1}^5 u_n$

M1

$$u_3 = 0.6u_2 (= 225) \text{ and } u_4 = 0.6^2u_2 (= 135)$$

Valid method to either find u_3 and u_4 or

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use of $\{S_n\} = \frac{a(1-r^n)}{1-r}$ *for either $n = 5$ or $n = 6$*

M1

$$\sum_{n=1}^5 u_n = 625 + 375 + 225 + 135 + 81 (= 1441)$$

A1

$$\sum_{n=6}^{\infty} u_n = 1562.5 - 1441 = 121.5$$

A1

4

Alternative:

Recognise that the sum to infinity with first term u_6 is required

(M1)

Valid method to find u_6 ($= 0.6 u_5$)

(M1)

$$\sum_{n=6}^{\infty} u_n = \frac{81 \times 0.6}{1 - 0.6}$$

(A1)

$$= 121.5$$

(A1)

[11]

Q10.

(a) $r = 16 \div 20 = 0.8$
OE

B1

1

(b) $\frac{a}{1-r} = \frac{20}{1-0.8}$

OE Using a correct formula with $a = 20$ or
 $r = c$'s 0.8

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$$= 100$$

ft on c's value of r provided $|r| < 1$

A1F

2

(c) $\{S_{20} = \frac{a(1-r^{20})}{1-r}$

OE Using a correct formula with $n = 20$

M1

$$= 100(1 - 0.8^{20}) = 98.847\{07..\}$$

Condone $> 3dp$

A1

2

(d) n th term $= 20 r^{n-1} = 20(0.8)^{n-1}$

Ft on c's r. Award even if 16^{n-1} seen

M1

$$= 20 \times 0.8^{-1} \times 0.8^n$$

$$= 25 \times 0.8^n$$

CSO; AG

A1

2

[7]

Q11.

(a) $ar = 48; \quad ar^3 = 3$

For either. OE

B1

$$\Rightarrow 16r^2 = 1$$

Elimination of a OE

M1

$$r^2 = \frac{1}{16} \Rightarrow r = -\frac{1}{4}$$

CSO AG Full valid completion.

SC Clear explicit verification (max B2 out of 3.)

A1

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or $r = \frac{1}{4}$

B1

4

(b) (i) $a = -192$

B1

1

(ii)
$$\frac{a}{1-r} = \frac{a}{1-\left(-\frac{1}{4}\right)}$$

$$\frac{a}{1-r} \text{ used}$$

M1

$$S_{\infty} = \frac{-768}{5} (= -153.6)$$

Ft on candidate's value for a. i.e. $\frac{4}{5}a$
SC candidate uses $r = 0.25$,
 gives $a = 192$ and sum to infinity = 256.
 (max. B0 M1A1)

A1ft

2

[7]

Q12.

(a) $u_1 = 12$

B1

$$u_2 = 3 \times 4^2 = 48$$

CSO AG (be convinced)

B1

2

(b) $r = 4$

B1

1

(c) (i) $\{S_{12} = \frac{a(1-r^{12})}{1-r}\}$

OE Using a correct formula with $n = 12$

M1

$$= \frac{12(1-4^{12})}{1-4}$$

Ft on answer for u_1 in (a) and r in (b)

A1ft

$$= \frac{12(1-4^{12})}{-3} = -4(1-4^{12}) = 4^{13} - 4$$

CAO Accept $k = 13$ for 4^{13} term

A1

3

$$\begin{aligned}
 \text{(ii)} \quad \sum_{n=2}^{12} u_n &= (4^{13} - 4) - u_1 \\
 &= 67108848
 \end{aligned}$$

B1

1

[7]

Q13.

(a) $\log 0.8^x = \log 0.05$ OR $x = \log_{0.8} 0.05$ **(M1)**

NMS: SC B2 for 13.425 or better
(B1 for 13.4 or 13.43; 13.42)

M1

$$x \log_{10} 0.8 = \log_{10} 0.05 \text{ oe}$$

A1

$$x = 13.425 \text{ to 3dp} \quad \text{OR} \quad 13.425 \text{ **(A2)**}$$

(else A1 for 1 or 2dp)

Condone greater accuracy

A1

3

(b) (i) $\frac{a}{1-r}$

$$S_{\infty} = \frac{a}{1-r} \text{ **used**}$$

M1

$$\frac{a}{1-r} = 5a \Rightarrow a = 5a(1-r)$$

Or better

A1

$$\Rightarrow 1 = 5(1-r) \Rightarrow r = \frac{4}{5} = 0.8$$

AG (be convinced)

A1

(ii) $n^{\text{th}} \text{ term} = 20 \times (0.8)^{n-1}$
Condone $20 \times (0.8)^{n-1}$

M1

$$n^{\text{th}} \text{ term} < 1 \Rightarrow 0.8^{n-1} < \frac{1}{20} \text{ oe}$$

$$0.8^{n-1} < 0.05 \text{ or } 0.8^{n-1} = k, \text{ where } k = 0.05$$

or k rounds **up** to 0.050

A1

Least n is 15

*If not 15, ft on integer part of [answer (a)+2]
 provided $n > 2$*

A1F

SC 3/3 for 15 if no error

SC $n^{\text{th}} \text{ term} = 16^{n-1}$ M1A0A0

3

[9]

Q14.

(a) $\frac{a}{1-r} = 4a$

(Accept $S_{\infty} = \frac{a}{1-\frac{3}{4}}$)

M1

$$\Rightarrow 1-r = \frac{a}{4a} \text{ or } a = 4a(1-r)$$

Either (or better) (or $S_{\infty} = 4a$ if M1 above)

A1

$$1-r = \frac{1}{4} \Rightarrow r = \frac{3}{4}$$

AG Be convinced. (or statement 4 times 1st term)

A1CSO

(b) $(S_{10} =) \frac{48(1-r^{10})}{1-r}$

Correct formula with $n = 10$ and one of

$$a = 48 \text{ or } r = \frac{3}{4} \text{ OE}$$

M1

$$= 192(1 - 0.75^{10}) = 181.1878 \text{ to 4dp}$$

A1

2

$$(c) \quad (i) \quad u_n = ar^{n-1} = a \left(\frac{3}{4}\right)^{n-1} = 48 \left(\frac{3}{4}\right)^{n-1}$$

B1

$$u_{2n} = ar^{2n-1} = a \left(\frac{3}{4}\right)^{2n-1} = 48 \left(\frac{3}{4}\right)^{2n-1}$$

ft on candidate's $u_n = ar^{\text{function of } n}$

B1ft

2

$$(ii) \quad \frac{u_n}{u_{2n}} = \frac{ar^{n-1}}{ar^{2n-1}} = \frac{r^{n-1}}{r^{2n-1}}$$

Eliminating a (or 48) or log a

M1

$$\log_{10} u_n - \log_{10} u_{2n} = \log_{10} \frac{u_n}{u_{2n}}$$

Using at least one log law

M1

$$= \log_{10} \frac{r^{n-1}}{r^{2n-1}} = \log_{10} (r^{-n})$$

$$= -n \log_{10} \frac{3}{4} = n \log_{10} \frac{4}{3}$$

A1CSO

AG Full valid completion

3

$$(iii) \quad \log_{10} \left[\frac{u_{100}}{u_{200}} \right] = 100 \log_{10} \left(\frac{4}{3} \right)$$

M1

= 12.49.... = 12.5 to 3 sf

AG Be convinced

SC: Those applying 'hence' to (i) rather than to (ii)

Mark as B2

A1CSO

2

[12]



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